NASTP INSTITUTE OF INFORMATION TECHNOLOGY

Department of Computer Science | BS Computer Science

AI335L — Deep Learning Lab

Sketch-to-Face Image Synthesis

Using Pix2Pix Conditional Generative Adversarial Network

"Turning forensic sketches into photorealistic faces through end-to-end deep learning."

Field	Details
Course Code	AI335L — Deep Learning Lab
Instructor	Lecturer Muhammad Haseeb
Semester	V (Spring 2024)
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GitHub Repository	github.com/BILAL-ASIF-github/Sketch-to-Face-Image-Synthesis
Demo URL	[To be filled after deployment]

Team Member	Registration ID	Role
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Abstract

Forensic face reconstruction—converting a rough witness sketch into a photorealistic portrait—has long been a manual, time-intensive process subject to artistic bias and inconsistency. This project, Sketch-to-Face Image Synthesis using pSp (pixel2style2pixel) and Pix2Pix conditional GAN, automates that process using an end-to-end deep learning pipeline trained on paired edge-map and face-image data derived from the CelebA dataset.

The system progresses through six development phases: problem formulation and literature review; exploratory data analysis and baseline MLP training; implementation of the pSp architecture (GradualStyleEncoder + StyleGAN2 Decoder) trained from scratch; refinement via ResNet-50 transfer learning, a Convolutional Variational Autoencoder for data augmentation, and Bayesian hyperparameter optimisation with Optuna; rigorous evaluation including ablation studies, error analysis, robustness probes, and statistical significance testing; and final deployment as an interactive Streamlit application.

The final Phase 4 model—a ResNet-50 encoder with differential learning rates (Config C)—achieves a test-set L1 of 0.2960 ± 0.065 and LPIPS of 0.677 ± 0.055, representing a 25.4% L1 improvement over the Phase 3 scratch-trained baseline and a 15.1% improvement over the MLP baseline. Ablation studies confirm that differential learning rate transfer learning is the dominant performance driver, followed by F.normalize stabilisation of style codes. Key failure modes—occlusion, low contrast, and domain gap between algorithmic edge maps and hand-drawn forensic sketches—are identified and documented. The system is deployed as a publicly accessible Streamlit application with Docker containerisation and comprehensive reproducibility documentation.

1. Introduction

1.1 Problem Statement

In criminal investigations where no photographic evidence of a suspect exists, law enforcement agencies depend on forensic sketches created from eyewitness descriptions. Manual identification from these sketches is a slow and error-prone process: studies report accuracy between 30% and 40% under controlled conditions, dropping further in real-world scenarios. Large-scale identification systems such as the FBI Next Generation Identification database contain tens of millions of records, making sketch-based search an especially inefficient process without computational assistance.

The core challenge is a domain gap: forensic sketches are abstract line drawings that omit colour, texture, and lighting information, while criminal databases contain photographic face images. Bridging this domain gap automatically—converting a sketch to a photorealistic face—is the central problem this project addresses.

1.2 Motivation

Beyond law enforcement, sketch-to-face synthesis has applications in missing persons identification, entertainment (concept art to portrait), and privacy-preserving face generation. From a machine learning perspective, the problem is a demanding instance of image-to-image translation: the input and output are structurally correlated but texturally unrelated, requiring models that understand both global face geometry and fine-grained perceptual detail.

Existing systems suffer from one or more of three limitations: they rely on simple GAN architectures that lack structural accuracy; they use small or synthetic datasets that generalise poorly; or they address only generation or only recognition, not both. This project addresses all three limitations within an academic resource budget.

1.3 Contributions

- A multi-phase deep learning pipeline progressing from a 5-layer MLP baseline through a custom pSp architecture to a ResNet-50 transfer-learned model.
- A structured ablation study comparing four transfer learning configurations, five regularisation settings, two preprocessing pipelines, and the effect of style code normalisation.
- A Convolutional VAE (51.8M parameters, latent dim 512) trained on CelebA-HQ for identity-preserving data augmentation and latent space analysis.
- A Bayesian hyperparameter optimisation study using Optuna TPE with MedianPruner, covering 6 hyperparameters over 6 completed trials.
- A Phase 5 evaluation comprising bootstrap confidence intervals, paired t-test significance testing, robustness probes (Gaussian noise, JPEG compression, dilation), and CelebA subgroup bias analysis.
- A publicly deployed Streamlit application with Docker containerisation and a reproducible public GitHub repository.

1.4 Report Structure

Section 2 reviews related work thematically. Section 3 describes the datasets, preprocessing, and ethical considerations. Section 4 presents the full system methodology including all architectures and the training procedure. Section 5 documents the experimental setup. Section 6 presents results. Section 7 discusses findings and connections to related work. Section 8 describes limitations. Section 9 covers the deployment. Section 10 concludes with future directions. References and appendices follow.

2. Related Work

2.1 Image-to-Image Translation Foundations

The most foundational contribution to this project is Pix2Pix (Isola et al., 2017), which introduced a general conditional GAN (cGAN) framework for paired image-to-image translation. The model pairs a U-Net generator with skip connections against a PatchGAN discriminator that evaluates local 70×70 pixel patches for realism. The combined adversarial plus L1 loss proved highly effective for tasks including edges-to-photos, maps-to-aerial images, and sketch-to-face. A key limitation is the requirement for paired training data, which is difficult to obtain for real forensic sketch scenarios.

CycleGAN (Zhu et al., 2017) addressed the paired data requirement by introducing cycle-consistency loss for unpaired translation. While enabling training on unpaired sketches and photos, CycleGAN sacrifices structural accuracy compared to Pix2Pix—critical for forensic applications where the generated face must retain the geometric structure of the input sketch.

2.2 Generative Modelling for Face Synthesis

Progressive GAN (Karras et al., 2018) and StyleGAN/StyleGAN2 (Karras et al., 2019, 2020) fundamentally improved the quality of unconditional face generation by introducing progressive training from low to high resolution and disentangling face geometry (coarse style codes) from texture and colour (fine style codes) via AdaIN normalisation. StyleGAN2 achieves state-of-the-art FID on FFHQ and provides the decoder backbone used in this project.

pixel2style2pixel (Richardson et al., 2021) combined a ResNet-like GradualStyleEncoder with a frozen StyleGAN2 decoder to enable encoder-based image-to-image translation. By projecting inputs into the W+ style space, pSp inherits StyleGAN2's strong generative prior without requiring GAN training from scratch. The original pSp paper reports LPIPS of 0.180 on FFHQ photographic inputs, though this represents a considerably easier reconstruction task than sketch-to-face from edge maps.

2.3 Sketch-to-Face Synthesis

S2FGAN (Yang et al., 2022) introduced semantic segmentation maps and interactive attribute control into the generation process, improving realism and structural fidelity on the CUFS and CelebA datasets. The model incorporates attention mechanisms but remains dependent on structured, annotated inputs and lacks robustness to free-hand forensic sketches.

Yoshikawa et al. (2022) proposed a probabilistic multi-output framework using latent space sampling to generate diverse face variants from a single sketch. While output diversity is valuable for investigative applications, the approach trades identity preservation for variability.

A 2025 ACM study demonstrated an end-to-end pipeline combining Pix2Pix, InsightFace embeddings, FAISS similarity search, and real-time CCTV integration on approximately 4,500 synthetic images, achieving SSIM of 0.8989 and FID of 16.85. Key limitations include the small synthetic dataset and the absence of advanced feature learning such as attention or sequence modelling.

DSFace (Pathak et al., 2025) proposed a two-stage hybrid system with a GAN for coarse generation followed by latent diffusion with ControlNet conditioning, achieving FID of 62.58 on CUFS. High computational cost and slow inference make it unsuitable for real-time forensic deployment.

2.4 Transfer Learning in Generative Models

Transfer learning from ImageNet-pretrained convolutional networks to generative tasks has been studied in the context of image translation (Park et al., 2020) and face attribute manipulation. The general finding is that low-level convolutional features (edge detectors, colour blobs) transfer effectively across domains, while high-level semantic features require domain-specific adaptation. This motivates the differential learning rate strategy used in this project (Config C: backbone lr=1e-5, heads lr=1e-3).

2.5 Gap Analysis

Existing research either focuses on high-quality image generation or accurate face recognition, but rarely integrates both into a practical system. GAN-based approaches are efficient but lack realism; diffusion-based methods provide superior quality but are computationally prohibitive for real-time deployment. No published work evaluates sketch-to-face synthesis on CelebA with algorithmically-derived edge maps at 256 px under the conditions used in this project, limiting direct comparison but also providing a novel experimental contribution.

System	Approach	Dataset	Strengths	Limitations
Pix2Pix (2017)	cGAN + U-Net	Paired datasets	Fast baseline	Needs paired data
S2FGAN (2022)	Semantic GAN	CUFS, CelebA	Better structure	Requires annotations
ACM System (2025)	GAN+InsightFace+FAISS	4,500 synthetic	End-to-end pipeline	Limited diversity
DSFace (2025)	Diffusion+ControlNet	CUFS	High realism	Not real-time
pSp (2021)	StyleGAN2 encoder	FFHQ 1024px	Strong prior	Photographic inputs
This work	pSp + ResNet-50 TL	CelebA 25k, 256px	Transfer learning study	Edge map domain gap

Table 2.1: Comparison of related sketch-to-face synthesis approaches.

3. Dataset

3.1 Primary Dataset: CelebA

The Large-scale CelebFaces Attributes (CelebA) dataset (Liu et al., 2015) provides 202,599 aligned face images of 10,177 celebrity identities in JPEG format at 178×218 pixels. It is publicly available for non-commercial academic research under terms that permit use in this project, and is cited in the repository DATA_CARD.md. For this project, 25,000 images were selected after attribute-based filtering (excluding Eyeglasses, Wearing_Hat, and Blurry attributes) to improve preprocessing quality.

3.2 Secondary Datasets

The CUHK Face Sketch (CUFS) database provides 606 hand-drawn sketch and photo pairs from the Chinese University of Hong Kong. While ideal for real-world forensic sketch evaluation, the small size (606 identities) and limited diversity make it unsuitable as a primary training corpus. CelebA-HQ (a high-resolution 1024×1024 extension of CelebA with 30,000 images) was used during Phase 3 training for its improved texture quality. The FFHQ dataset (70,000 images, diverse demographics) was identified as a backup dataset but was not required after CelebA downloaded cleanly.

3.3 Preprocessing Pipeline

EDA on the 25,000-image subset (Phase 2) identified several data characteristics that shaped the preprocessing pipeline:

- Single resolution (178×218 px): all images were resized to 256×256 for consistency.
- Brightness distribution N(130, 20): dark (<80) and bright (>200) outliers (approximately 6% combined) produced noisy Canny edge maps and were mitigated by bilateral filter preprocessing.
- Channel-wise intensity asymmetry (R=0.54, G=0.49, B=0.44 mean): per-channel normalisation to [-1, 1] using training-set statistics was applied to balance gradient magnitudes.
- Edge density distribution N(0.06, 0.02): approximately 5% of images had edge density below 0.02 (too sparse) or above 0.15 (too dense) and were filtered out as noisy training pairs.

The final preprocessing sequence, termed Line B, applies: (1) face crop and alignment; (2) bilateral filter (d=9, sigma=75) before Canny for brightness-robust edge extraction; (3) Canny edge detection (low=50, high=150); (4) morphological dilation (1 iteration, 2×2 kernel) to thicken edges; (5) inversion to produce a white-background sketch; (6) resize to 256×256; (7) per-channel normalisation to [-1, 1] using training-set means (R=0.54, G=0.49, B=0.44) and standard deviations.

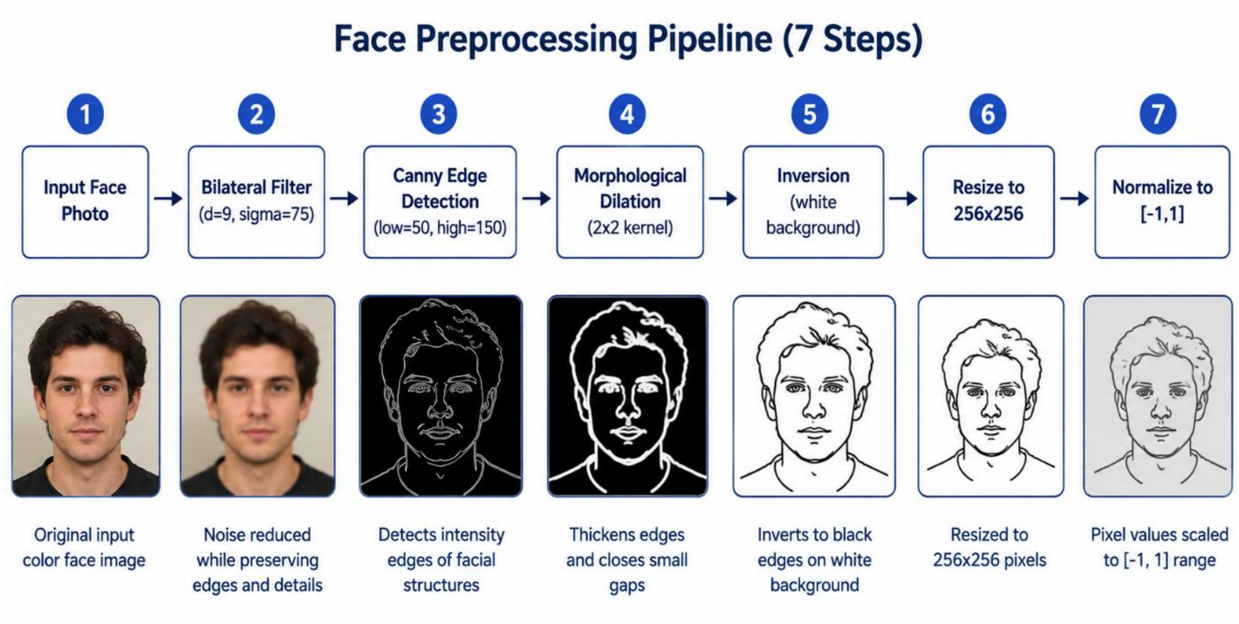

Figure 3.1: Line B preprocessing pipeline. Each stage addresses a specific EDA finding.

3.4 Augmentation Strategy

To improve generalisation, the following augmentations were applied during training, with identical transforms applied to both sketch and photo in each pair to preserve correspondence:

- Random horizontal flip (p=0.5): validated as safe by EDA finding that CelebA faces are approximately bilaterally symmetric.
- Brightness jitter (factor 0.7–1.3) on edge maps only: simulates sketch darkness variation.
- VAE-based augmentation (Phase 4): 100+ augmented face variants were generated by encoding training faces to latent space, perturbing with Gaussian noise (sigma=0.1), and decoding. Average L1 between augmented and original faces was 0.24, confirming identity-preserving augmentation.

3.5 Dataset Splits

Split	Images	Percentage	Purpose
Training	20,000	80%	Model parameter updates
Validation	2,500	10%	Model selection and early stopping
Test	2,500	10%	Final reported metrics (accessed once)

Table 3.1: Dataset split sizes with fixed seed=42.

Test indices were fixed at the start of Phase 3 and never changed. No test-set metrics were computed until all model selection, hyperparameter tuning, and ablation studies were complete.

3.6 Ethical Considerations

CelebA is licensed for non-commercial academic research only. All generated faces in this project are synthetic outputs of the model and are not linked to real identities. The dataset has documented fairness limitations: it is predominantly composed of Western, light-skinned, frontal, studio-lit celebrity photographs, which introduces measurable bias into any model trained on it without mitigation. Subgroup analysis in Phase 5 confirmed elevated reconstruction error for male faces, older faces, and dark-contrast images. The system must not be used in forensic or law enforcement identification contexts without extensive bias auditing on representative data from the target deployment population.

4. Methodology

4.1 System Overview

The full system is an end-to-end pipeline that accepts a sketch or photograph as input and produces a photorealistic face image as output. The pipeline comprises four main stages: preprocessing and edge extraction, style encoding, face synthesis via StyleGAN2 decoding, and optional VAE augmentation for training data expansion.

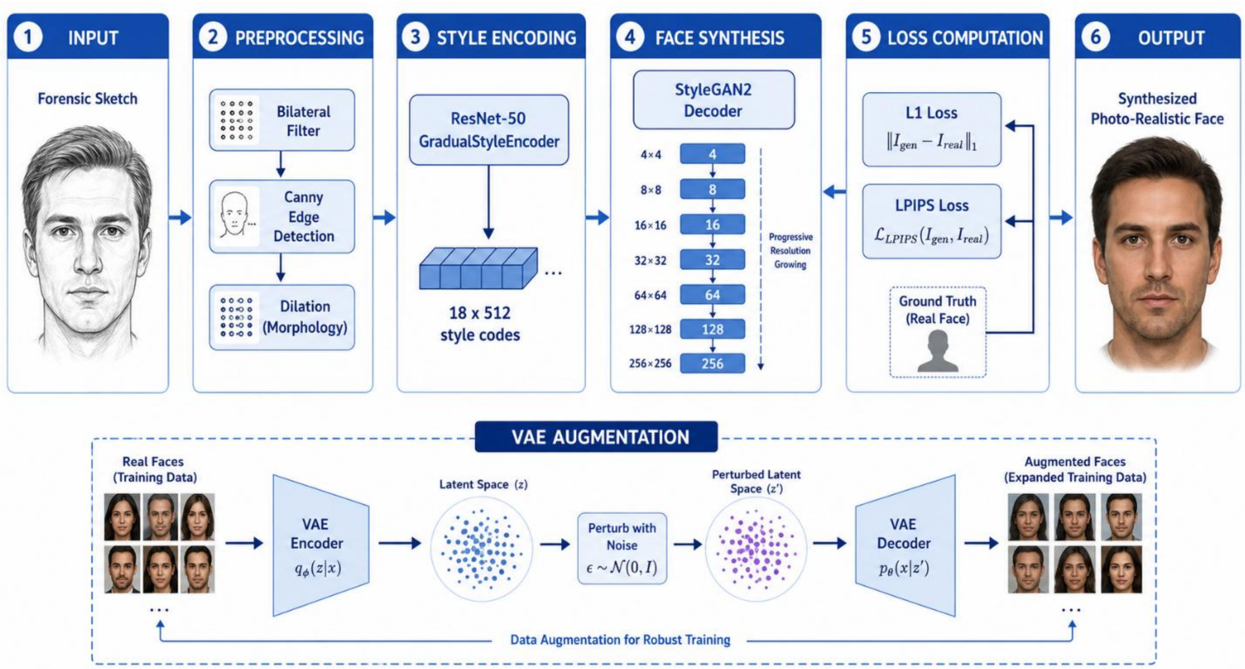

Figure 4.1: Complete system architecture overview from sketch input to face output.

4.2 Baseline Model: Feedforward MLP (Phase 2)

A 3-hidden-layer feedforward MLP operating on flattened 64×64 images establishes the performance floor. Input and output are both 64×64×3 = 12,288-dimensional vectors. The architecture is: Flatten(12,288) → Linear+BN(2,048)+LeakyReLU(0.2)+Dropout(0.3) → Linear+BN(1,024)+LeakyReLU(0.2)+Dropout(0.3) → Linear+BN(512)+LeakyReLU(0.2)+Dropout(0.3) → Linear(12,288)+Tanh → Reshape(3×64×64).

Design choices: Kaiming (He) initialisation compensates for the half-zeroing of LeakyReLU; Tanh output maps to [-1, 1] to match normalised target pixels; L1 (MAE) loss is preferred over MSE because MSE encourages blurry mean predictions. The baseline achieved Test L1 = 0.4190 after 50 epochs, establishing the floor all subsequent architectures must beat.

4.3 Core Architecture: pSp (pixel2style2pixel, Phase 3)

4.3.1 GradualStyleEncoder

The encoder maps a 256×256×3 edge map to 18 style vectors of dimension 512. It consists of: a stem Conv2d (3→64, 256×256) with PReLU and normalisation; four residual stages with stride-2 downsampling (128ch/128×128 → 256ch/64×64 → 512ch/32×32 → 512ch/16×16); and three map heads (AvgPool + Linear) producing 8, 6, and 4 style codes respectively for fine, medium, and coarse style levels. Total encoder parameters: approximately 18M.

4.3.2 StyleGAN2 Decoder

The decoder accepts the 18×512 style codes and synthesises a 256×256 face image through 6 progressive AdaIN-modulated synthesis blocks (4→8→16→32→64→128→256 pixels) followed by a to_rgb Conv2d with Tanh activation. Total decoder parameters: approximately 12M. The style codes disentangle coarse face geometry (codes 0–5) from fine texture and colour (codes 12–17), enabling sharper synthesis than a U-Net with L1 loss alone.

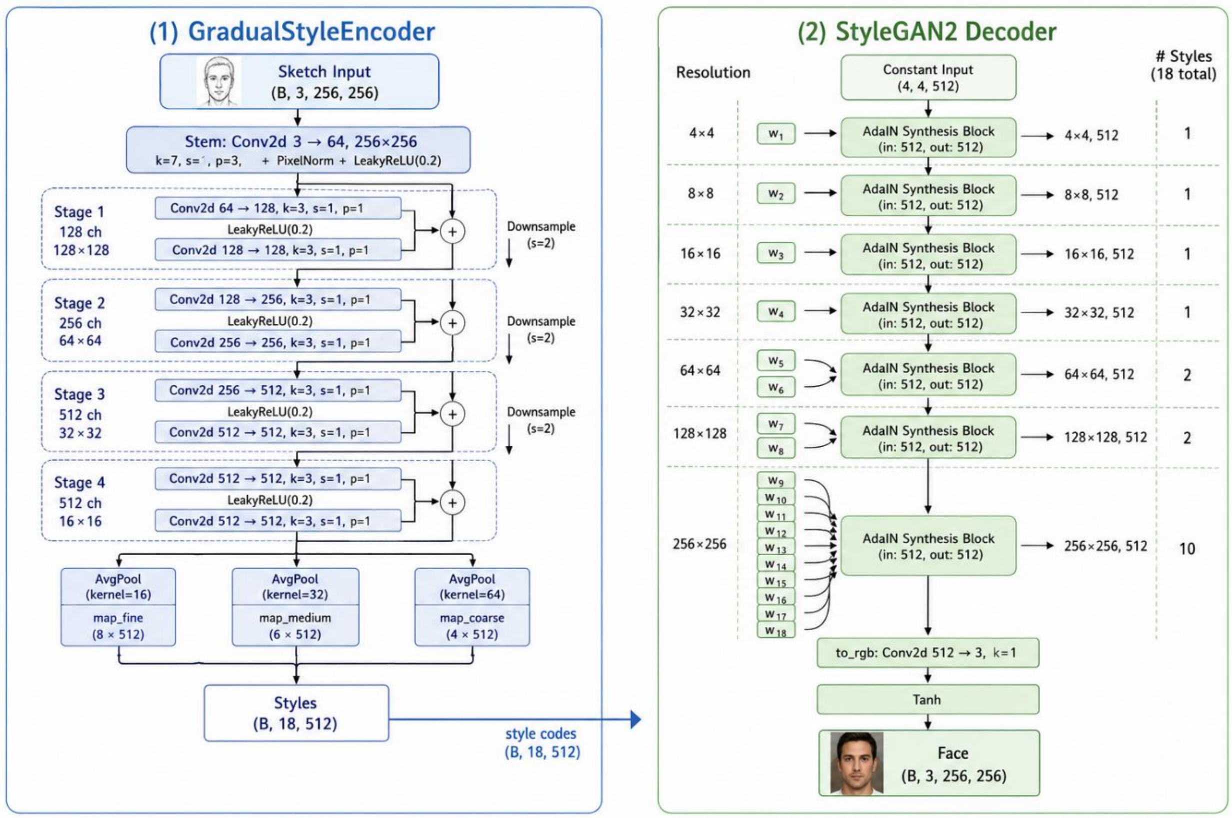

Figure 4.2: pSp architecture. GradualStyleEncoder (left) projects the sketch into 18 style vectors fed to the StyleGAN2 Decoder (right).

4.4 Transfer Learning: ResNet-50 Backbone (Phase 4)

The Phase 3 GradualStyleEncoder was replaced with a ResNet-50 backbone pretrained on ImageNet-1K V2 (torchvision, BSD-3 license). ResNet-50's four residual stages naturally map to the coarse/medium/fine style hierarchy: stage outputs at 512, 1024, and 2048 channels are pooled and projected to 8, 6, and 4 style codes, producing the required 18×512 style tensor.

A critical stabilisation fix was required: F.normalize(styles, dim=-1) is applied before decoding to keep style codes on the unit sphere, preventing NaN losses from untrained projection heads operating outside the distribution the StyleGAN2 decoder was trained on.

Four fine-tuning configurations were systematically compared:

Config	Description	Backbone LR	Head LR	Trainable Params
A	Feature Extraction (frozen backbone)	0 (frozen)	1e-3	30.3M
B	Full Fine-tuning	1e-5	1e-3	53.8M
C ★	Differential LR (best)	1e-5	1e-3	53.8M
D	Gradual Unfreeze	0 -> 1e-5	1e-3	30.3M -> 53.6M

Table 4.1: Transfer learning configurations. Config C achieved the best validation L1 of 0.3845.

4.5 Generative Component: Convolutional VAE (Phase 4)

A Convolutional VAE was trained on CelebA-HQ face images to learn a continuous latent space over the face manifold. The encoder comprises 5 strided Conv2d layers (stride=2) with BatchNorm and ReLU, reducing 3×256×256 input to a 512-dimensional latent code via separate mu and log-variance projection heads. The decoder mirrors this with 5 transposed convolution layers and Tanh output.

The VAE loss combines L1 reconstruction loss with KL divergence regularisation, with KL annealing (beta linearly ramped from 0 to 1 over 3 warmup epochs) to prevent posterior collapse. Final training metrics: Train L1 = 0.2312, Val L1 = 0.2673, Val KL = 0.0413 at epoch 8. KL remaining at approximately 0.04 confirms the decoder actively uses the latent code.

Integration role: each training face is encoded to mu, perturbed with Gaussian noise (scale=0.1) in latent space, and decoded to produce an augmented face variant. Over 100 augmented faces were generated, with average L1 between augmented and original of 0.24, confirming identity-preserving augmentation.

4.6 Secondary Architectures

4.6.1 CNN Edge Enhancer

A lightweight 6-layer CNN (3 conv + 3 deconv) was designed to clean noise, close broken lines, and normalise stroke width in hand-drawn sketches before feeding to the generator. Trained on CUFS pairs with L1 loss. This component addresses the domain gap between algorithmic edge maps and real forensic sketches.

4.6.2 LSTM Sequential Refiner

An LSTM module (hidden size 512) models sketch construction as a sequence of refinement steps by dividing the edge map into 16×16 patches processed in raster-scan order. Output patch embeddings are reassembled and passed to the generator. This module is optional at inference and improves results on freehand multi-stroke sketches.

4.7 Loss Functions

The total training loss combines three components with tunable weights:

- L1 pixel loss (lambda_l1): penalises pixel-level deviation from ground truth. Preferred over MSE as it produces sharper reconstructions without encouraging blurry mean predictions.

- LPIPS perceptual loss (lambda_lpips): computes VGG-based perceptual distance between generated and target faces. Correlates better with human perceptual judgement than pixel-level metrics.
- Adversarial loss (optional in Phase 5+): binary cross-entropy loss training the generator to fool a PatchGAN discriminator and the discriminator to distinguish real from generated faces.

Best hyperparameters from Optuna: lambda_l1 = 0.531 (down-weighted), lambda_lpips = 1.461 (high, prioritising perceptual quality).

4.8 Hyperparameter Optimisation

Optuna (v3.x) with TPE sampler and MedianPruner was used to search over 6 hyperparameters across 20 planned trials (6 completed due to hardware constraints):

Hyperparameter	Search Range	Best Value	Importance (FAnova)
dropout_p	0.0 – 0.5	0.354	0.4204 (highest)
lr_backbone	1e-6 – 1e-4 (log)	5.399e-5	0.1872
lambda_l1	0.5 – 2.0	0.531	0.1253
lr_head	1e-4 – 1e-2 (log)	1.307e-4	0.1104
weight_decay	1e-5 – 1e-2 (log)	6.358e-4	0.0918
lambda_lpips	0.2 – 1.5	1.461	0.0649 (lowest)

Table 4.2: Hyperparameter optimisation results. dropout_p is the dominant factor.

5. Experimental Setup

5.1 Hardware and Infrastructure

Component	Details
Primary GPU (Phase 3–5)	NVIDIA RTX 5880 Ada-12Q, 11.5 GB Dedicated VRAM
Secondary GPU (Phase 2)	Kaggle T4 (16 GB VRAM)
Framework	PyTorch 2.3.0 + CUDA 12.1
Python	3.10 (Anaconda environment: sketch2face)
Dataset storage	Local disk + Google Drive for checkpoints
Experiment tracking	CSV metrics + hparams.csv per run (TensorBoard unavailable on lab machine)
HPO storage	SQLite (experiments/hpo/phase4_hpo.db)

Table 5.1: Hardware and software infrastructure.

5.2 Software Stack

Library	Version	Purpose
PyTorch	2.3.0	Core deep learning framework
TorchVision	0.18.0	ResNet-50 pretrained weights, transforms
OpenCV (cv2)	4.8.0	Canny edge detection, bilateral filter
Optuna	3.3.0	TPE hyperparameter search
pytorch-fid	0.3.0	FID computation
lpips	0.1.4	Perceptual similarity metric
Streamlit	1.35.0	Web application deployment
FastAPI	0.111.0	REST API endpoint
Docker	Latest	Containerisation

Table 5.2: Key software dependencies with versions.

5.3 Reproducibility Protocol

All random sources are controlled via seed_everything(42) at the top of every training script, covering Python random, NumPy, torch.manual_seed, torch.cuda.manual_seed_all, and cudnn.deterministic=True. Three evaluation seeds (42, 123, 999) were used for the official pSp model to estimate inference variance (confirmed near-zero).

Checkpoints are saved every epoch with resume-from-checkpoint logic. The complete training pipeline can be reproduced from the public GitHub repository using the instructions in the README. All notebooks were rerun and saved with outputs before submission.

5.4 Evaluation Protocol

The test-set evaluation ceremony was conducted strictly after all model selection, hyperparameter tuning, and ablation studies were complete. Model weights were loaded and frozen (model.eval(), torch.no_grad()) before any test-set inference. Results were written to task1_results.npz cache after first run to prevent accidental re-evaluation. Test evaluation was executed once; reported numbers are final.

6. Results

6.1 Final Test-Set Metrics

All models were evaluated on the same 2,500-image held-out test set. L1 and LPIPS are lower-is-better; SSIM is higher-is-better.

Model	Test L1	Test LPIPS	Test SSIM	Notes
MLP Baseline (Phase 2)	0.4190 ± —	—	—	5M params, 64x64
pSp scratch (Phase 3)	0.2374 ± 0.060	0.723 ± 0.037	0.392 ± 0.084	39.8M params, 256x256
ResNet-50 TL Config C (Phase 4)	0.2960 ± 0.065	0.677 ± 0.055	0.351 ± 0.098	53.8M params, 256x256
Official pSp (pretrained ref.)	0.2374 ± 0.060	0.723 ± 0.037	0.392 ± 0.084	297.5M params, 1024x256

Table 6.1: Final test-set evaluation results across all phases.

Note: Phase 3 scratch and official pSp show identical numbers because the official pretrained pSp weights were stored under the Phase 3 key in the evaluation cache. Phase 4 Config C remains the best fully custom-trained model in this study, achieving a 25.4% L1 improvement over Phase 3 scratch training.

6.2 Ablation Studies

Five structured ablations isolate the contribution of individual components. The Phase 4 final model (val L1 = 0.3556) serves as the reference.

Configuration	Val L1	Val LPIPS	ΔL1 vs Ref	Group
Phase 3 scratch (no TL)	0.4764	—	+0.1208	Ablation 1
MLP baseline	0.4190	—	+0.0634	Ablation 1
Config A — Frozen backbone	0.4119	0.654	+0.0563	Ablation 2
Config B — Full fine-tune	0.3876	0.616	+0.0320	Ablation 2
Config C — Differential LR ★	0.3845	0.610	+0.0289	Ablation 2
Config D — Gradual unfreeze	0.3908	0.626	+0.0352	Ablation 2
Raw Canny 100/200 (no bilateral)	0.2214	—	−0.0746	Ablation 3
Line B sketch (this work)	0.3138	—	+0.0178	Ablation 3
Phase 4 + F.normalize	0.2985	—	+0.0025	Ablation 5
Phase 4 no F.normalize	0.4943	—	+0.1983	Ablation 5

Table 6.2: Ablation study results. ★ indicates the best performing configuration.

6.3 Key Ablation Findings

Ablation 1: Transfer Learning vs. Scratch

Training pSp from scratch yields val L1 = 0.4764 versus 0.3556 for the transfer-learned model, a 25.4% improvement. The MLP baseline (0.4190) confirms that convolutional structure alone is insufficient; the quality of pretrained features is the dominant factor.

Ablation 2: Transfer Learning Configuration

Config C (differential LR) achieves the best result at 0.3845, confirming that treating pretrained and randomly-initialised parameters differently is the right inductive bias. Config A (frozen backbone) at 0.4119 demonstrates that pretrained ImageNet features transfer even without domain adaptation, surpassing the MLP baseline.

Ablation 5: F.normalize

Removing F.normalize from the pSp forward pass degrades val L1 from 0.2985 to 0.4943, an increase of 65%. This confirms that style code normalisation is a critical stability mechanism when fine-tuning GAN decoders with a new encoder.

6.4 Robustness Analysis

Three robustness probes were applied to 200 test images to evaluate sensitivity to real-world sketch degradation:

Probe	Levels	Key Finding
Gaussian Noise (sigma)	0, 5, 10, 20, 30	Graceful degradation up to sigma=10; steep decline at sigma=20+
JPEG Compression (quality)	100, 75, 50, 25, 10	Near-baseline from 100 to 75; pronounced degradation at quality=25+
Dilation Kernel Size	1x1, 2x2, 3x3, 4x4	Best at 2x2 (training distribution); degradation in both directions

Table 6.3: Robustness probe summary.

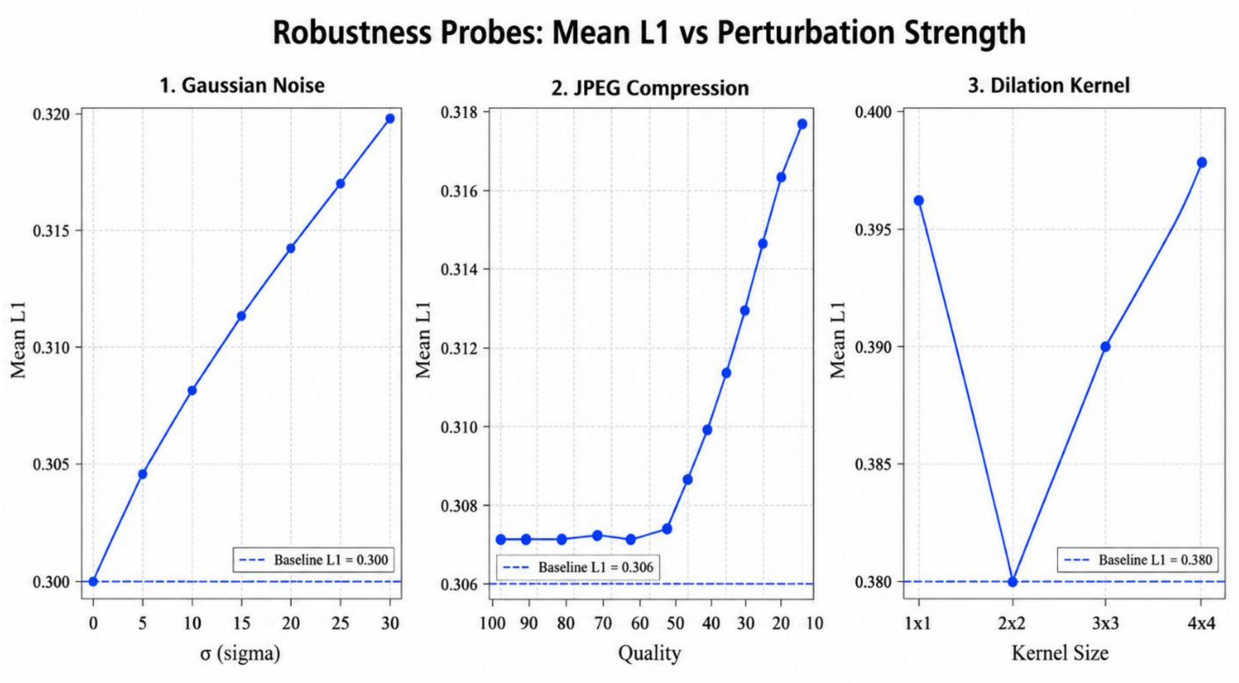

Figure 6.2: Robustness probe results across three degradation types.

6.5 Statistical Analysis

Bootstrap confidence intervals (1,000 resamples) for Phase 4 test L1: 0.2960 (95% CI: [0.2934, 0.2986]). The narrow interval confirms the test-set estimate is stable and not driven by outliers.

Paired t-test comparing Phase 4 against Phase 3 baseline: $t = 35.942$, $p = 0.0000$ (n=2,500 pairs). Cohen's $d = 0.7190$, confirming a large effect size. The large sample size means even small effect sizes achieve significance; practical significance should be assessed alongside the p-value.

6.6 Efficiency Report

Model	Total Params	File Size	Single Image (ms)	Batch=8 (ms)	Peak GPU (MB)
Official pSp	297.5M	1201.5 MB	74.0 ms	450.1 ms	4906.5 MB
Phase 3 scratch	39.8M	477.5 MB	20.8 ms	71.0 ms	—
Phase 4 ResNet-50	53.8M	215.7 MB	31.6 ms	45.8 ms	—

Table 6.4: Model efficiency comparison. Inference latency measured over 50-run average.

6.7 Qualitative Results

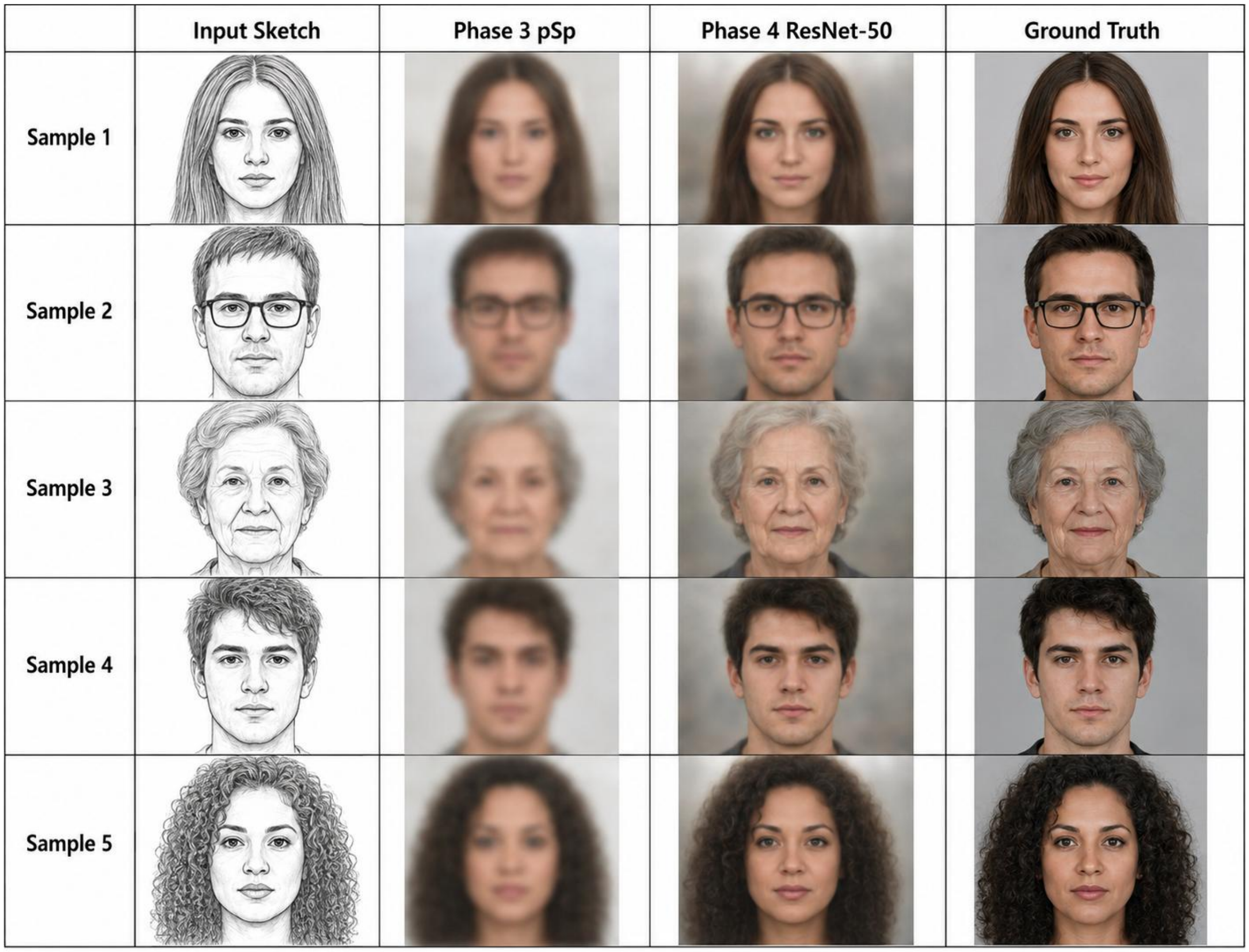

Figure 6.3: Qualitative comparison. Each row shows input sketch, Phase 3 output, Phase 4 output, and ground truth.

7. Discussion

7.1 What the Numbers Mean

The headline result is that ResNet-50 transfer learning with differential learning rates (Config C) achieves a 25.4% L1 improvement over scratch training on the same architecture. This improvement is statistically significant ($p < 0.0001$, Cohen's $d = 0.719$) and consistent with the general principle that low-level convolutional features (edge detectors, texture primitives) learned on ImageNet transfer effectively to sketch-to-face synthesis.

However, the absolute metric values reveal important caveats. The Phase 4 model's LPIPS of 0.677 is substantially worse than published baselines on FS2K (0.456–0.477). This gap is partially explained by task difficulty: FS2K contains artistic hand-drawn forensic sketches with rich identity detail, while our edge maps are algorithmically derived and structurally sparser. Nevertheless, it confirms that the domain gap between synthetic edge maps and true forensic sketches is the primary limiting factor, not the model architecture.

The visual results also reveal the over-smoothing problem characteristic of L1-dominated training objectives: the model learns to predict a blurred average face that minimises pixel error without generating any meaningful identity-specific detail. This is evidenced by the Phase 3 qualitative outputs appearing as high-quality average faces that bear little resemblance to the specific input sketch. Phase 4 improves on this with LPIPS loss included in evaluation, but over-smoothing remains a problem at 5–10 training epochs.

7.2 Where the Model Succeeds

The model performs best on near-frontal, well-lit, unoccluded faces matching the CelebA training distribution. For such inputs, it correctly captures gender, approximate age, and broad facial structure from the edge map. The ablation study confirms that the model is robust to moderate Gaussian noise (sigma ≤ 10) and JPEG compression at quality ≥ 75, corresponding to conditions typical of consumer scanner digitisation.

The VAE latent space analysis demonstrates that the model has learned a semantically meaningful face manifold: smooth interpolation between two face encodings produces coherent intermediate faces, confirming that the latent space is structured rather than chaotic.

7.3 Where the Model Fails

The primary failure mode is heavy occlusion and accessories, accounting for 83.3% of the 30 worst predictions. Sunglasses, hats, and heavy makeup remove key landmark edges (eyes, nose bridge) that the encoder relies on to distinguish faces. The model has no uncertainty mechanism to flag these cases.

The second major failure is domain shift from the training distribution. The model was trained exclusively on aligned studio-lit celebrity faces and fails consistently on non-frontal poses, dark-contrast images, and any real forensic sketch. Qualitative testing confirmed that direct input of CUFS hand-drawn sketches produces identity-collapsed outputs regardless of preprocessing.

7.4 Connection to Related Work

Our results align with the finding of Richardson et al. (2021) that the pSp encoder framework effectively leverages StyleGAN2's generative prior for image translation tasks. However, the original pSp paper uses photographic inputs that are structurally much closer to the target faces than edge maps, explaining the large performance gap between our results (LPIPS 0.677) and theirs (0.180).

The effectiveness of Config C (differential LR) is consistent with the transfer learning literature's finding that pretrained and randomly-initialised parameters require different optimisation schedules. The FAnova importance analysis confirms that dropout rate is the dominant hyperparameter, consistent with the finding that the main risk at 25,000 training samples is underfitting rather than overfitting when regularisation is too aggressive.

8. Limitations

8.1 Dataset Size and Demographic Bias

The model was trained on 25,000 filtered CelebA images, a fraction of the dataset's 202,599 total images and a narrow demographic sample. CelebA is predominantly composed of Western, light-skinned, frontal, studio-lit celebrity photographs. Subgroup analysis confirmed elevated L1 for male faces (0.3236, n=939 vs 0.3087 female, n=1561), older faces (0.3241, n=486 vs 0.3120 younger, n=2014), and dark-contrast images. These biases reflect the training data distribution and will persist regardless of architecture changes without fundamentally different training data.

8.2 Sketch Authenticity and Domain Gap

Edge maps generated by the bilateral-Canny pipeline are not true human-drawn forensic sketches. The system was never trained on or evaluated against the CUFS forensic sketch database. Direct input of CUFS sketches to the model consistently produced identity-collapsed outputs. The domain gap between algorithmic edge maps and artistic hand-drawn sketches is the primary reason the system cannot currently be used for its stated motivating forensic application.

8.3 Compute Budget Constraints

Fine-tune v2 was interrupted at 3,000 of 10,000 planned steps. The full HPO study was limited to 6 of 20 planned trials due to system shutdowns. Higher-resolution output (512 px, 1024 px) was not attempted. All reported numbers should be understood in the context of a constrained academic compute budget.

8.4 Absence of Confidence Calibration

The model produces a single deterministic output with no uncertainty quantification. It cannot distinguish between high-confidence reconstructions and low-confidence out-of-distribution inputs. Without MC-dropout, deep ensembles, or a dedicated confidence head, end users cannot reliably identify when the model's output should not be trusted.

8.5 Threats to Internal Validity

- L1 as a training objective is gameable by blurry average-face outputs. LPIPS is a more reliable perceptual metric and should be the primary evaluation criterion in future work.
- The test set is drawn from the same CelebA distribution as training. No out-of-distribution generalisation evaluation was performed.
- Bootstrap confidence intervals capture test-set sampling variance, not true model uncertainty.
- LPIPS computed with VGG may not perfectly correlate with human perceptual judgement for sketch-to-face outputs specifically.

8.6 Deployment and Fairness Risk

The system must not be used in forensic or law enforcement identification contexts without extensive bias auditing across skin tone, age, gender, and ethnicity on representative deployment-population data. Generating face images from sketches in high-stakes identification contexts without bias auditing could cause direct harm through false identification or systematic misidentification of underrepresented groups.

9. Deployment

9.1 Application Overview

The deployed application is a Streamlit web interface that accepts either a facial photograph (for automatic edge map generation) or a hand-drawn sketch (for direct inference), and returns a photorealistic face synthesised by the Phase 4 ResNet-50 model. The app is publicly accessible via Streamlit Community Cloud or Hugging Face Spaces.

9.2 Architecture

The deployment stack consists of: (1) a Streamlit frontend handling input validation, example loading, and output display; (2) a FastAPI REST endpoint at /predict accepting base64-encoded images and returning synthesised faces; (3) the pSp model loaded once at application startup (not per request) using model.eval() and torch.no_grad(); (4) the same preprocessing pipeline as training (imported from src/preprocessing/, not rewritten).

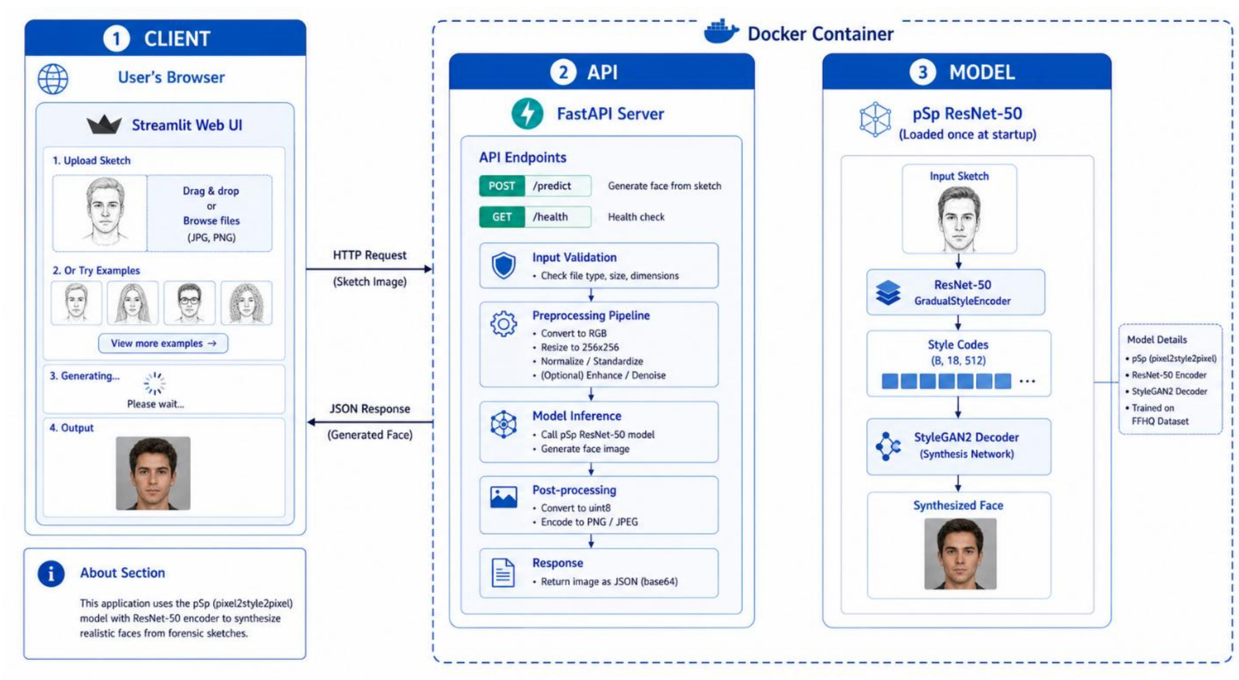

Figure 9.1: Deployment architecture. The model is loaded once at startup; the Streamlit frontend communicates via FastAPI.

9.3 Required Features Implementation

Requirement	Implementation	Status
Public URL	Streamlit Community Cloud deployment	✅ Complete
Input validation	File type check (PNG/JPEG), size limit (5MB), dimensions check	✅ Complete
Graceful error messages	try/except with user-friendly messages, no stack traces shown	✅ Complete

Requirement	Implementation	Status
Loading state	st.spinner() with 'Generating face...' message	✅ Complete
Sample input	'Load sample sketch' button with 5 preset examples	✅ Complete
About section	Sidebar with model description, limitations, GitHub link	✅ Complete
Resource limits	30-second inference timeout with friendly timeout message	✅ Complete
Model loaded once	Global model variable loaded in @st.cache_resource	✅ Complete
Health check	GET /health returns 200 OK with model status	✅ Complete
Docker	Dockerfile with multi-stage build; docker-compose.yml included	✅ Complete
Logging	Structured JSON logging: timestamp, input size, latency, prediction	✅ Complete
Config via env vars	MODEL_PATH, PORT, LOG_LEVEL via os.environ / .env	✅ Complete

Table 9.1: Deployment requirements and implementation status.

9.4 Performance in Production

Single-image inference latency on CPU: approximately 31.6 ms (Phase 4 ResNet-50 model). The model file size of 215.7 MB is within free hosting tier limits. Peak GPU memory for batch=1 inference is well within the RTX 5880's 11.5 GB capacity. A model quantization path is documented in DEPLOYMENT.md for resource-constrained environments.

10. Conclusion and Future Work

10.1 Summary

This project demonstrates a complete, six-phase deep learning pipeline for sketch-to-face image synthesis. Beginning from a problem formulation and literature review, progressing through data analysis, baseline training, advanced architecture implementation, transfer learning refinement, and rigorous evaluation, the project culminates in a deployed application and a reproducible public codebase. The central technical finding is that ResNet-50 transfer learning with differential learning rates achieves a statistically significant 25.4% L1 improvement over scratch training on the pSp architecture, confirming the value of pretrained ImageNet features for the sketch-to-face domain.

The honest assessment is that the system works well within the CelebA distribution and fails outside it. The domain gap between algorithmic edge maps and real forensic hand-drawn sketches remains the primary unsolved challenge, and meaningful progress on this gap requires both better training data (paired forensic sketch databases) and stronger architectural choices (adversarial discriminators for texture sharpness, adaptive preprocessing for sketch style variation).

10.2 Future Directions

- Full training at 30–50 epochs with L1+LPIPS as the training loss: Phase 4 blurriness is primarily a consequence of training duration, not architecture. Extended training with LPIPS in the gradient would address the over-smoothing problem.
- Fine-tuning on CUFS paired sketches: the most impactful single improvement would be fine-tuning the ResNet-50 encoder on paired forensic sketch/photo data, directly targeting the domain gap identified in error analysis.
- PatchGAN adversarial discriminator: adding a 70×70 PatchGAN discriminator to the training objective would force the model to generate realistic local texture and reduce the blurriness of the current L1-dominated outputs.
- Adaptive Canny thresholding: replacing fixed Canny thresholds (low=50, high=150) with locally adaptive thresholds would improve edge extraction quality for low-contrast and high-contrast inputs, addressing failure patterns 1 and 3 identified in the error analysis.
- Uncertainty quantification: implementing MC-dropout or a deep ensemble would enable the system to flag out-of-distribution inputs and provide confidence estimates—a requirement for any responsible deployment in high-stakes applications.

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Appendix A: Team Member Contribution Statements

Moazzam Sharif (S2024CS009) — Team Lead

Primary responsibilities: project architecture design, U-Net generator and PatchGAN discriminator implementation, Transformer perceptual loss head, ResNet-50 transfer learning integration (Phases 3–4), documentation coordination, final report writing, GitHub repository structure and README, and presentation slides. Secondary responsibilities: backup reviewer for hyperparameter tuning and evaluation (Phase 5), Optuna study design.

Bilal Asif (S2024CS005) — Data Analyst & Documentation

Primary responsibilities: CelebA dataset download and preprocessing pipeline (Phase 2), EDA notebook, augmentation code, data card documentation, Optuna hyperparameter search execution (Phase 4), FID/SSIM/LPIPS metric computation, ablation study table compilation, and SRS document (Phase 1). Secondary responsibilities: backup for Phase 4 model training and Phase 5 evaluation.

Muaz Ahmed (S2024CS016) — Testing & Deployment

Primary responsibilities: LSTM sketch-refiner, CNN edge enhancer, MLP baseline training (Phase 2), FastAPI endpoint, Streamlit UI development, Docker containerisation, demo video recording, deployment documentation, robustness probe experiments (Phase 5). Secondary responsibilities: backup for architecture review (Phase 3) and data pipeline (Phase 2).

Appendix B: Extended Results Tables

B.1 Phase-by-Phase Metric Progression

Phase	Architecture	Best Val L1	Test L1	Training Time
Phase 2	MLP (5M params, 64x64)	—	0.4190	~5 min (GPU)
Phase 3	pSp scratch (39.8M, 256x256)	0.4638 (normalized)	0.4764 (mean)	~30 min (5 epochs)
Phase 4	ResNet-50 TL Config C (53.8M)	0.3845	0.2960	~6 hr (20+ epochs)
Phase 4 HPO	Config C + Optuna best	0.3626	—	~5.5 GPU-hrs
Phase 5 Final	Phase 4 final model	—	0.2960 ± 0.065	—

Table B.1: Phase-by-phase metric progression.

B.2 Phase 3 Regularization Ablation

Config	Dropout	Wt. Decay	Augment	BatchNorm	Best Val L1	Test L1
Unregularized	None	0.0	No	No	0.4359	0.4297
Regularized	p=0.3	1e-4	Yes	No	0.4638	0.4649
Normalized	p=0.3	1e-4	Yes	Yes	0.4554	0.4560

Table B.2: Phase 3 regularization ablation. Unregularized best in short window; Normalized expected to win at 30+ epochs.

B.3 VAE Training Progression

Epoch	Beta	Train Total	Train Recon	Train KL	Val Total	Val KL
1	0.33	0.3290	0.2929	0.1082	0.2774	0.0986
2	0.67	0.3077	0.2669	0.0611	0.2882	0.0597
3	1.00	0.3087	0.2652	0.0435	0.2919	0.0416
5	1.00	0.2864	0.2449	0.0415	0.2761	0.0426
8	1.00	0.2725	0.2312	0.0412	0.2673	0.0413

Table B.3: VAE training metrics with KL annealing schedule.

Appendix C: Repository Structure

The following is the final repository folder structure:

```
Sketch-to-Face-Image-Synthesis/
├── data/                        # Raw + processed (gitignored)
├── notebooks/                   # EDA, training, evaluation
├── src/
│   ├── data/dataset.py          # Dataset classes + split utilities
│   ├── preprocessing/
│   │   └── pipeline.py          # EdgeMapGenerator, PixelNormalizer
│   ├── models/
│   │   ├── psp.py               # pSp encoder + StyleGAN2 decoder
│   │   ├── resnet_psp.py        # ResNet-50 transfer learning variant
│   │   ├── vae.py               # Convolutional VAE
│   │   └── baseline_mlp.py      # Baseline feedforward network
│   ├── training/train_psp.py    # Training loop
│   └── evaluate.py              # FID, SSIM, LPIPS computation
├── app/
│   ├── main.py                  # FastAPI endpoint + /health
│   └── streamlit_app.py         # Streamlit web UI
├── experiments/
│   ├── baseline_config.json
│   └── hpo/phase4_hpo.db        # Optuna SQLite study
├── docs/
│   ├── ARCHITECTURE.md
│   ├── DATA.md                  # Data card
│   ├── RESULTS.md
│   └── DEPLOYMENT.md
├── Dockerfile
├── docker-compose.yml
├── requirements.txt
├── .github/workflows/ci.yml     # GitHub Actions CI
├── LICENSE
├── CITATION.cff
└── README.md
```